

Marcus Winter

SOFTWARE ENGINEER

M.S. Computer Science, Brown University

+1 (401) 537 9663 ■ marcus.a.winter@gmail.com



SUMMARY

Seeking career in software engineering post master's degree completion in May 2026.

Passionate about High quality digital experiences, Game development, Machine learning, and Computer graphics.

PROJECTS AND SKILLS

For a full list and in-depth information, check out my website! <https://mwinter02.github.io/>

- Custom data-oriented Game engine | [info](#) | C++, OpenGL, GLSL, CMake, Game Dev, Data structures
- Full-stack AI powered web app | [info](#) | Java, React, Firebase, Full-Stack, Team programming
- Attribute trained urban scene GAN | [info](#) | Python, TensorFlow, Machine Learning, VCS (Git)
- Cross platform sports recruiting app | [info](#) | Dart, Flutter, Full-Stack, UI/UX, Mobile development

WORK EXPERIENCE

Course Developer, Brown University C.S. Department

October 2025 – February 2026

- Developed course content for CSCI 1950U – “3D Game Engine Development” where students create a 3D game engine in C++. Topics include ECS, physics, AI and pathfinding, data structures, etc.
- Created a stencil renderer in C++ utilizing OpenGL for students to build their engine upon, also created homework assignments such as a 3D collision lab to visualize translation vectors for GJK/EPA collision.
- Worked collaboratively with the course professor and teaching assistants to address problems in previous offerings to ensure a smooth learning experience.

Head Teaching Assistant, Brown University

August 2025 – January 2026

- Taught CSCI 1950N – “2D Game Engines” – a high-level CS course where students create their own 2D game engine in Java and multiple games during a 12-week semester.
- Worked collaboratively with a fellow TA to deliver weekly lectures, host office hours, and manage course content delivery. The professor gave us full responsibility for the course and gave minimal input.
- Topics included UI management, component-based entity system, sprite rendering, collisions, accurate physics, behavior tree AI, pathfinding, spatial partitioning, and more.

Executive Intern, TerraCycle Inc

June 2024 – December 2024

- Worked with the executive team to deliver an internal tool for analytics and marketing.
- Developed a visual dashboard utilizing Microsoft Power BI which pulls data from SQL database.
- Worked on site and remotely, performing weekly progress updates and a final product presentation to the CEO and CAO.

Research Intern, Sandfield NZ

June 2023 – December 2023

- Worked on a project for a client utilizing MATLAB for data management and presentation.
- Performed research on SQL databases and AWS integration solutions for future work.

EDUCATION

- | | | |
|--|------------------|---------------------------|
| ▪ M.S. Computer Science | Brown University | September 2024 – May 2026 |
| ▪ B.S. Applied Math & Computer Science | Brown University | September 2021 – May 2025 |